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cfDNA fragmentation patterns correlate with tumor burden measured via PSMA PET/CT volumetric parameters in patients with biochemical recurrence of prostate cancer

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Abstract

Background Prostate cancer recurrence following primary treatment poses a significant clinical challenge, particularly when detected through biochemical recurrence at low PSA levels. Conventional imaging modalities often fail to localize the disease at this early stage. PSMA PET has demonstrated superior sensitivity in detecting recurrent lesions, even in patients with low PSA. Concurrently, liquid biopsy, through analysis of cell-free DNA (cfDNA), offers a minimally invasive approach for monitoring disease. There is scarce evidence about the association between liquid biopsy and PSMA PET/CT findings. This study aimed to assess the correlation between liquid biopsy and tumor burden assessed by PSMA PET/CT in early recurring prostate cancer patients.

Results PSMA PET/CT and liquid biopsies of 32 patients in biochemical recurrence were analyzed. 12 patients (37.5%) had no PSMA PET-measurable disease. Four patients (12.5%) presented local recurrence, seven (21.9%) had recurrence in pelvic lymph nodes, one of whom also had local recurrence. Nine patients (28.1%) presented metastatic recurrence, with or without local or nodal recurrence. PSA levels correlated with molecular imaging data ($p < 0.05$), including whole body PSMA-TV, whole body PSMA-TL, whole body SUVmean and whole body SUVmax. The mean cfDNA fragment size fraction was inversely correlated with tumour burden measured with whole body PSMA-TV, with a Spearman correlation coefficient of -0.451 and a p-value of 0.009. No correlation was found between cfDNA concentration and PET-PSMA data.

Conclusion This prospective study demonstrated a statistically significant negative correlation between cfDNA fragmentation patterns and PSMA PET/CT volumetric parameters in patients with presumed localized prostate cancer

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with early biochemical recurrence. These findings underscore the potential of liquid biopsy as a biomarker and a complementary tool to PSMA PET/CT to assess disease progression during the follow-up of these patients.

Keywords Prostate cancer, Biochemical recurrence, PSMA PET, Liquid biopsy

Introduction

Prostate cancer (PCa) is the second most common cancer in men worldwide and the most frequent malignancy in Europe and the USA. It had an estimated global mortality of 375,000 in 2020 [1]. Serum prostate-specific antigen (PSA) quantification, imaging techniques such as magnetic resonance imaging (MRI), and prostate biopsy are the current tools for PCa diagnosis [2]. Usually, treatment of localized PCa patients includes a local approach like radical prostatectomy (RP) or radiotherapy [3]. After treatment, serial PSA quantification is used to detect biochemical recurrence (BR). Overall, within two years of RP, approximately 14% of overall PCa patients experience BR [4], and this percentage is higher in patients with high-risk factors [5, 6]. Despite BR being detected, not all patients present detectable disease recurrence with current imaging techniques such as computed tomography (CT) and bone scintigraphy.

Prostate-specific membrane antigen (PSMA) is highly expressed in PCa cells, [7] making it an excellent target for diagnostic imaging. PSMA PET has demonstrated superior diagnostic accuracy and a greater impact on management decisions [8] compared with conventional methods. Particularly during the early stages of BR, both anatomical imaging and bone scintigraphy frequently fail to accurately identify the site of relapse, while PSMA PET has proven useful in detecting recurrent lesions, even in patients with low PSA levels [9, 10]. PSMA PET can pinpoint the exact location of recurrence, enabling localized treatments such as targeted radiotherapy or salvage surgery [11]. Thus, PSMA PET has emerged as the preferred imaging tool for diagnosis in the setting of BR after radiotherapy treatment [12–14] and is recommended after RP, as indicated in the EAU guidelines [15, 16], especially at low PSA levels [17, 18]. Assessing tumour volume (denoted as PSMA-TV) is advised during PSMA PET interpretation, as growing evidence suggests that it may be a prognostic factor of overall survival [19].

Alternatively, liquid biopsy techniques to study molecular biomarkers in several cancer types, including PCa are gaining considerable importance. Liquid biopsy analysis provides cancer-specific information from a simple and minimally invasive blood extraction that can be repeatedly obtained from patients, making it an ideal approach for disease monitoring, during both diagnosis and follow-up [20]. One of the components found in liquid biopsy samples is cell-free DNA (cfDNA), which

consists of highly fragmented nucleic acids secreted from apoptotic cells, both healthy and cancer cells. High cfDNA levels are associated with poor prognosis in different urologic tumours, including PCa [21–23]. It seems that cfDNA secreted by tumour cells is shorter than that originating from non-malignant cells [24–26]. Moreover, cfDNA fragmentation patterns can provide diagnostic and prognostic value.

There is scarce evidence about the association between liquid biopsy and PSMA PET/CT findings [27–29], however, there are promising results. To the best of our knowledge, the correlation between PSMA PET and liquid biopsy in early recurring PCa patients has not been studied yet. Therefore, this study aimed to assess the correlation between cfDNA concentration, and mean cfDNA fragment fraction and tumour burden assessed by PSMA PET in the setting of BR PCa patients.

Material and methods

Patients

Patients referred to the Nuclear Medicine department for PSMA PET/CT in the setting of BR of PCa between February 2022 and February 2023 were prospectively included in the study. None of the patients included had other active neoplasms. All patients had previously undergone RP. BR was defined as PSA > 0.4 ng/ml and rising as per EAU guidelines [30, 31]. As a PSMA PET/CT performance criterion is PSA > 0.2 ng/ml, we also included 10 patients with PSA levels ranging from 0.2 to 0.4 ng/ml. The clinical and pathological characteristics of these patients are shown in Table 1. All BR patients were re-staged with PSMA PET/CT scans using [¹⁸F]F-DCFPyL in 29 patients (91%) or [⁶⁸Ga]Ga-PSMA-11 in three (9%). The median PSA level during recurrence was 0.74 ng/mL (IQR 0.94).

One 10 mL EDTA tube of peripheral blood was collected when PET PSMA was performed due to BR and stored at 4°C until processed within the following four hours. Peripheral blood was then used to isolate cfDNA.

PSMA PET/CT acquisition protocol

PSMA PET/CT was performed using a Biograph mCT TrueV PET/CT hybrid device (Siemens, Germany), with low-dose CT for attenuation correction and image fusion. The radiotracers used were [¹⁸F]F-DCFPyL and [⁶⁸Ga]Ga-PSMA-11, at doses of 4 and 2 MBq/Kg,

Table 1 Demographic and clinicopathological data of the cohort studied

Characteristics	Cohort patients studied (n = 32)
Age of diagnosis, median (range)	63 (45–76)
Pathologies of interest, n (%)	
None	12 (37.5)
DM	3 (9.4)
HT	15 (46.9)
Dyslipidemia	11 (34.4)
Initial PSA ng/mL, median (range)	7.8 (2.04–38.4)
Prostatic volume cc, median (range)	35 (15–98)
ISUP score, n (%)	
1	3 (9.4)
2	6 (18.75)
3	11 (34.4)
4	6 (18.75)
5	5 (15.6)
Pathological Stage, n (%) ^a	
pT2	12 (37.5)
pT3a	9 (28.1)
pT3b	9 (28.1)
pN, n (%)	
X	7 (22)
0	18 (56.25)
1	6 (18.75)
Number of previous recurrences, n (%)	
0	20 (60)
1	10 (33.3)
2	2 (6.6)

PCa prostate cancer, DM diabetes mellitus, HT hypertension, PSA prostate-specific antigen, ISUP International Society for Urological Pathology

^a The pathological stage of one patient was not available from medical records

respectively. PSMA PET/CT imaging was acquired from the skull base to the proximal third of the thigh with arms raised above the head, 60 min after radiotracer injection for [⁶⁸Ga]Ga-PSMA-11 and 90 min p.i. for fluorine tracer, and at 3 min and 2.5 min per bed. PET data reconstruction was performed using a standard iterative algorithm from CT records.

Image analysis and interpretation

PSMA PET/CT images were assessed using a workstation (syngo.PET&CT Oncology VA20A, Siemens Healthineers AG, Germany) by a physician in training and a senior nuclear medicine specialist experienced in PSMA PET/CT. Any discrepancies were resolved by consensus. Pathological findings were defined following the Second Version of the Prostate Cancer Molecular Imaging Standardized Evaluation Framework Including Response

Evaluation for Clinical Trials (PROMISEv2) criteria [30]. Spherical or ellipsoidal regions of interest were placed over all pathological uptakes on PSMA PET/CT images, ensuring that the entire lesion was enclosed in axial, sagittal, and coronal projections. Standardized uptake values (SUVs) (SUVmax and SUVmean) were automatically calculated based on measured activity concentration (Bq/mL) multiplied by patient weight (kg) and normalized to injected activity (Bq).

To obtain PSMA-derived tumour volume values (PSMA-TV), the contouring margins of each lesion were delineated by a threshold of 40% SUVmax using an SUV-based automated contouring program (Syngo.via, Siemens Healthineers, Germany). Volumetric whole-body analysis was performed to assess the PSMA tumour burden, including all lesions: local recurrence, lymph nodes, and distant metastases, which together constitute the whole-body PSMA tumour burden. The sum of the PSMA-TV of each lesion was defined as whole-body (wb) PSMA-TV (wbPSMA-TV), which was recorded and calculated for the entire disease.

To report the extent and location of recurrence lesions in PSMA PET images, PROMISEv2 criteria were used. Patients were classified into subgroups based on the presence of local, nodal, and/or metastatic disease according to a combination of the following categories: no local tumour (T0), local recurrence (Tr), with no positive pelvic lymph nodes (N0), lymph node metastases in a single pelvic lymph node region (N1), lymph node metastases in more than one pelvic lymph node region (N2), no distant metastases (M0), lesions in distant lymph node regions (M1a), bone metastases (M1b), and metastases in other sites (M1c).

cfDNA isolation and quantification

To separate plasma, blood samples were centrifuged at 3500 rpm for 15 min at 4°C, followed by plasma centrifugation at 16,000×g for 10 min at 4°C to remove any remaining cells. Plasma samples were then stored at -80°C until cfDNA extraction.

cfDNA was extracted from 1.5 to 4 mL of plasma (depending on availability) using the QIAamp Circulating Nucleic Acid Kit (Qiagen, Hilden, Germany), according to the manufacturer's instructions. Plasma cfDNA concentration was obtained using the Quant-it PicoGreen dsDNA Assay kit. The mean cfDNA fragment size fraction was evaluated using the Agilent 2200 TapeStation System. The average size fraction from short fragments (100–250 bp) was calculated from the electropherogram.

Statistical analysis

PSMA PET/CT data were compared against cfDNA levels and mean fragmentation patterns in the cohort

studied using Spearman's rank correlation coefficient. Moreover, PSMA PET/CT and liquid biopsy data were correlated with clinicopathological variables from each patient including PSA levels using Spearman's rank correlation coefficient. Statistical significance was established at a p-value of 0.05. All analyses were carried out with the SPSS software package (IMB SPSS Statistics 25).

All participants provided written informed consent (HCB/2013/8753) before being included in this study. The study methodology conformed to the standard set by the Declaration of Helsinki and was approved by

the Clinical Research Ethics Committee of the Hospital Clínic Barcelona (HCB/2022/0542).

Results

Of the 32 patients with BR included, 12 (37.5%) had no PSMA PET/CT-measurable disease. Four patients (12.5%) presented local recurrence and seven (21.9%) had a recurrence in the pelvic lymph nodes, one of whom also had local recurrence. Nine patients (28.1%) presented metastatic recurrence, with or without local or nodal recurrence (Fig. 1). The recurrence sites with their

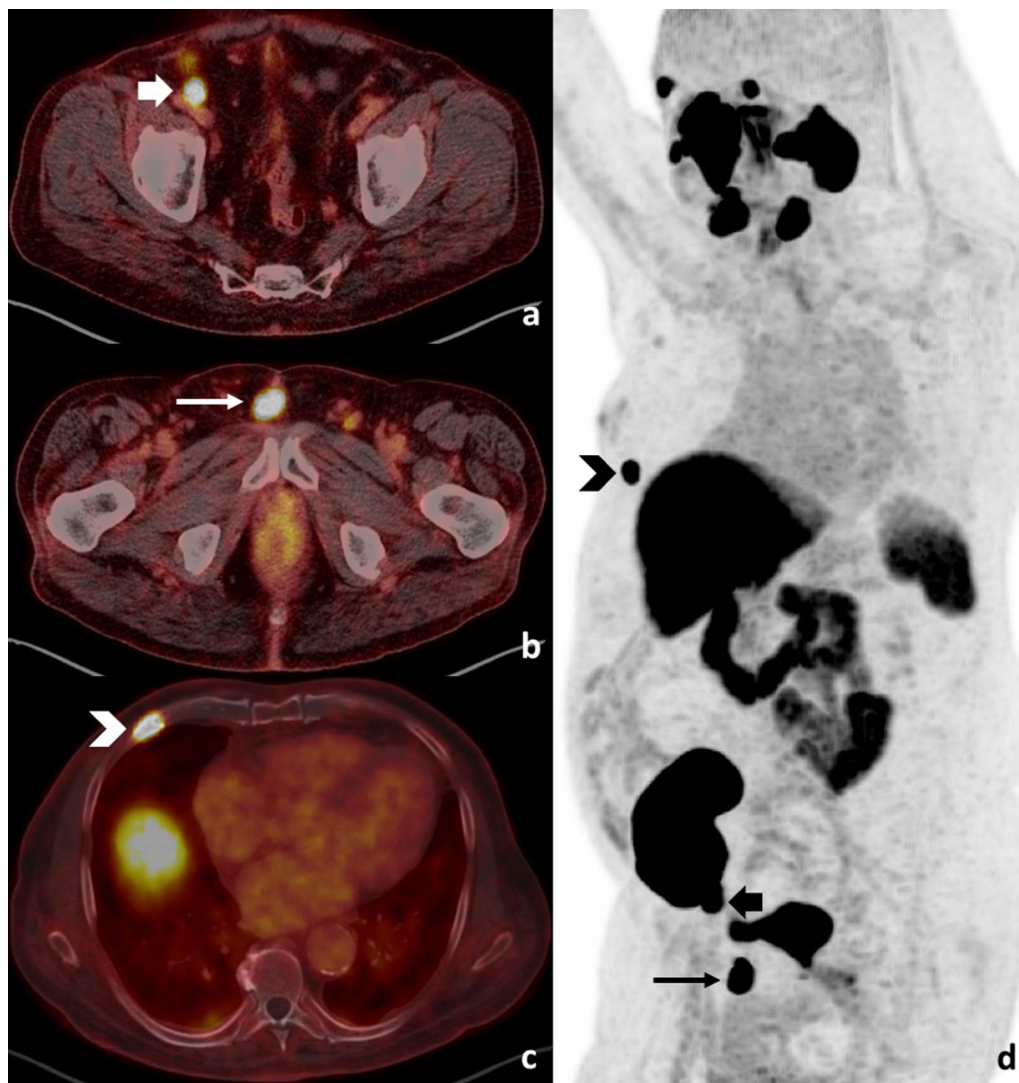


Fig. 1 PSMA PET/CT showing metastatic sites of biochemical recurrence PSMA PET/CT performed with [^{18}F]F-DCFPyL in a 68 year-old patient with biochemical recurrence. Fused PET/CT axial images (**a**, **b**, **c**) and MIP image (**d**) revealed a pelvic right external iliac node (thick arrow), a distant lymph node at the prepubic area (thin arrow), and a bone metastasis in a right rib (arrowhead), corresponding to a T0N1M1b stage according to PROMISE v2 criteria. The patient presented with a wbMTV40 of 6.37 cc and a wbTLG40 of 149.95. The cfDNA level was 4.42 ng/mL with a mean fragment size of 133 bp. The patient had a renal graft in the right iliac fossa

Table 2 Patients classified according to recurrence sites with their corresponding staging

Recurrence site		Staging	n	n total
No local, nodal or metastatic recurrence		T0N0M0	12	12
Local recurrence		TrN0M0	4	4
Pelvic nodal recurrence	Only nodal	T0N1M0	6	7
	Local + nodal	TrN2M0	1	
Metastatic recurrence	Distant nodes only	T0N0M1a	1	9
	Bone metastasis only	T0N0M1b	2	
	Local + Distant nodes	TrN0M1a	1	
	Pelvic nodes + Distant nodes	T0N1M1a	1	
	Pelvic nodes + Bone metastasis	T0N1M1b	1	
	Pelvic nodes (1) + Distant nodes + Bone metastasis	T0N1M1b	1	
	Pelvic nodes (> 1) + Distant nodes + Bone metastasis	T0N2M1b	2	

corresponding staging according to PROMISEv2 criteria are detailed in Table 2.

PSA levels correlated with molecular imaging data, including wbPSMA-TV40 (Spearman's correlation coefficient (SCC)=0.489; $p=0.005$), wbTL-PSMA40 (SCC=0.449; $p=0.01$), wbSUVmean (SCC=0.418; $p=0.017$) and wbSUVmax (SCC=0.428; $p=0.014$) (Fig. 2).

PSMA PET/CT parameters—such as wbPSMA-TV, wbTL-PSMA40, wbTL-PSMA40, wbSUVmax and wbSUVmean— and liquid biopsy parameters—such as cfDNA concentration and mean cfDNA fragmentation patterns—in recurrent patients based on disease extension are shown in Table 3.

The mean cfDNA fragment size fraction was inversely correlated with tumour burden measured via wbPSMA-TV, with an SCC of -0.451 and a p -value of 0.009

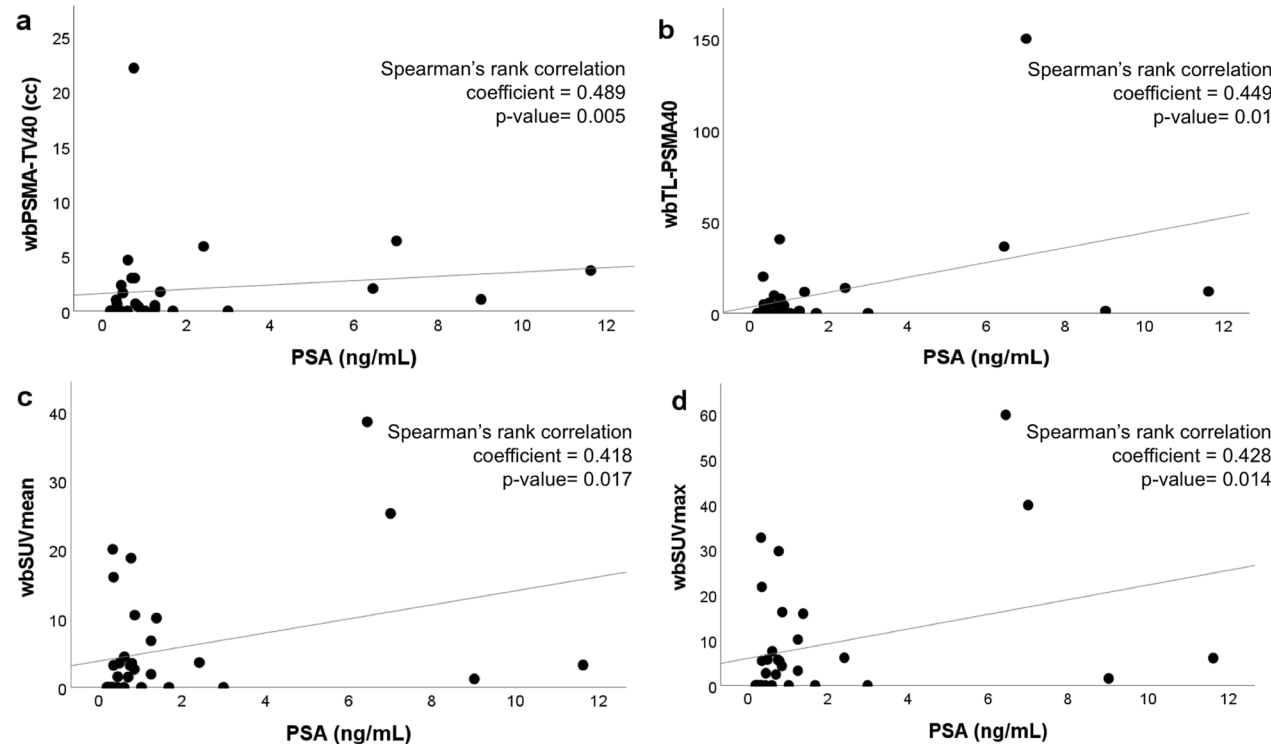
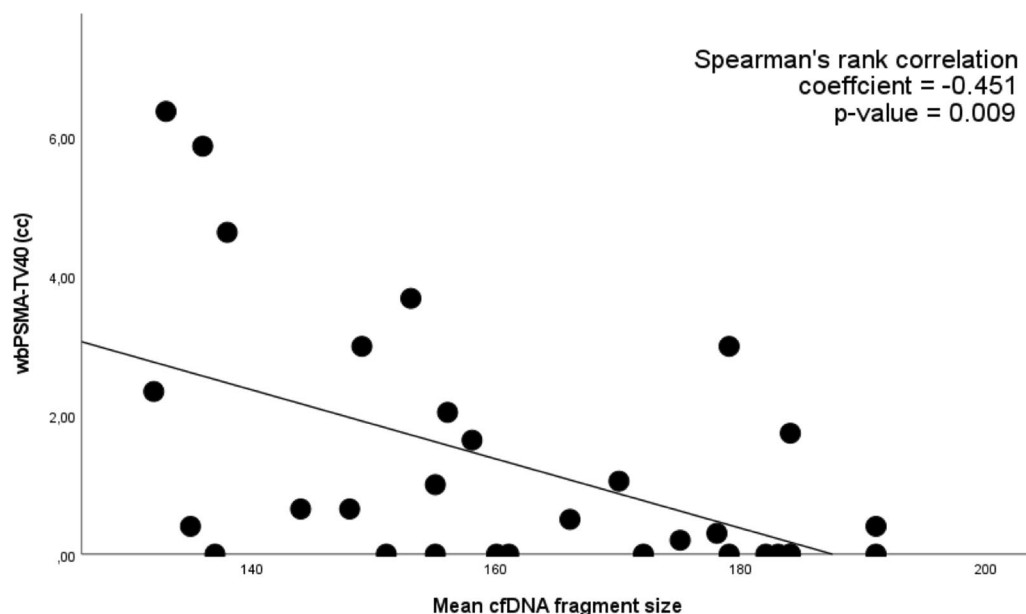


Fig. 2 PSA level correlates with molecular imaging data, including wbPSMA-TV40 (a), wbTL-PSMA40 (b), wbSUVmean (c) and wbSUVmax (d)

Table 3 PSA, wbPSMA-TV, TL-PSMA40, cfDNA concentration and mean cfDNA fragment size fraction for extension of disease; median (IQR)

Extension of disease	PSA (ng/mL)	wbPSMA-TV40 (cc)	wbTL-PSMA40 (cc)	wbSUVmax	wbSUVmean	cfDNA concentration (ng/mL)	Mean cfDNA fragmentation
Total cohort (n=32)	0.74 (0.94)	0.45 (2.12)	1.70 (8.47)	3.77 (8.19)	2.25 (5.00)	11.03 (7.46)	160 (30.25)
No measurable disease (TONOMO) (n=12)	0.33 (0.7)	0	0	0	0	11.76 (8.27)	175.5 (27.5)
Local recurrence (Tr) (n=4)	0.58 (3.92)	0.65 (0.85)	3.52 (4.35)	5.73 (9.96)	3.35 (3.39)	11.53 (1.9)	150.5 (26.75)
Pelvic nodal disease (N1, N2) (n=7)	0.79 (0.80)	1.37 (2.22)	3.58 (5.79)	4.31 (5.87)	2.61 (3.88)	10.3 (7.24)	149 (40)
Distant nodal disease (M1a) (n=3)	1.39 (2.78)	1.74 (0.82)	11.60 (16.09)	16.7 (21.95)	10.48 (14.28)	9.75 (4)	184 (17.5)
Bone metastasis (M1b) (n=6)	1.6 (6.82)	4.43 (4.27)	10.92 (27.38)	5.91 (18.08)	3.55 (11.72)	7.35 (5.9)	161.5 (37)

**Fig. 3** Mean cfDNA fragment size fraction correlates with tumour burden measured with wbPSMA-TV

(Fig. 3). No correlation was found between cfDNA concentration and PSMA PET/CT data.

Discussion

After local treatment, between 25 and 50% of PCa patients will develop BR and be at an increased risk of developing metastasis. Imaging can help to determine whether the recurrence is local or distant; however, its performance is PSA-dependent. Liquid biopsy may be useful in combination with PSMA PET to improve metastasis assessment.

We assessed the correlation between cfDNA concentration and mean cfDNA fragment size fraction and

image-derived volumetric parameters from PSMA PET/CT in a cohort of PCa patients with RP at the time of BR. To our knowledge, this is the first study focused on PSMA PET and liquid biopsy in early recurring PCa patients. We found a statistically significant negative correlation between wbPSMA-TV and mean cfDNA fragment size fraction. Short cfDNA fragments are associated with apoptosis of cancerous cells, implying that these patients might have a higher tumour burden and a higher risk of progression to metastatic disease. This result is consistent with other studies conducted to date in different settings [27–29, 31]. Chen et al. compared cfDNA in healthy controls (n=34), patients with localized PCa

($n=112$), and patients with metastatic castration-resistant prostate cancer (mCRPC) ($n=122$). They found that localized PCa patients had a shorter average cfDNA fragment size when compared with controls [26]. Similar correlations have been observed in other malignancies. In a previous study on stage III-IV non-small cell lung cancer (NSCLC) patients ($n=53$) [29]; cfDNA fragmentation correlated with fludeoxyglucose-18 (FDG) PET/CT parameters from primary tumour, extrapulmonary disease and whole-body disease. In this sense, mean cfDNA fragmentation patterns in combination with PSMA PET parameters might be useful to detect patients with a high risk of progression.

On the other hand, we found no correlation between cfDNA levels and volumetric parameters from PSMA PET/CT. In addition, there were no differences in cfDNA levels between patients with different recurrence sites classified using PSMA PET. Kluge et al. compared the correlation of liquid biopsy data and PSMA PET between patients with hormone-sensitive prostate cancer (HSPC) ($n=74$) and mCRPC ($n=74$). They found that cfDNA levels were weakly correlated with PSMA-TV in the mCRPC group but not in the HSPC group [31]. Chen et al. observed that plasma cfDNA concentrations were significantly elevated in patients with mCRPC but not in patients with localized disease. Along that same line, a study from González de Aledo-Castillo et al. on NSCLC found a positive correlation between cfDNA concentration and extrapulmonary FDG PET/CT volumetric parameters, while no association was found between cfDNA concentration and FDG PET/CT parameters from local disease. The previously published results indicated a stronger correlation in advanced disease, which account for the absence of correlation in our series.

It is important to note that, as illustrated in the cohort studied in this work, BR is a highly heterogeneous entity that includes patients with local, nodal or oligometastatic extension. Our cohort included patients with early BR, with low tumour burden. Regarding liquid biopsy and PSMA PET parameters, this group of patients might behave similarly to those with localized disease in previously described studies—which also found no significant differences in cfDNA concentration. Furthermore, it could differ from cohorts with more advanced stages of the disease, specifically in patients with mCRPC, in whom higher levels of cfDNA were found. Considering this, cfDNA concentration may identify patients with high tumour burden but it might present limitations when distinguishing between healthy individuals and patients with localized PCa or early BR.

PSMA PET/CT may produce false negatives in patients with micrometastases and oligometastases. Studies by Budaus et al. [32] and Van Leeuwen et al.

[33] comparing presurgical PSMA PET/CT with histopathological findings of removed lymph nodes reported a 33% and 64% sensitivity for lymph node metastasis detection, respectively. Both studies found that false-negative lesions were significantly lower than true-positives. This shows that lesion size impacts PSMA PET detection, which is a crucial matter in early recurrence with a small tumour burden. This could have altered the correlation between PSMA PET/CT and liquid biopsy parameters, especially considering that cfDNA is a non-tumour-specific marker that, despite its utility and availability, can be affected by other factors, such as tissue damage or chronic diseases. Despite this, we found a statistically significant correlation between liquid biopsy and PSMA PET/CT, consistent with previous studies on the subject.

To the best of our knowledge, this is the first work to focus specifically on patients with BR. We acknowledge the limitations of our study, mainly the limited sample size that might have affected the statistical analysis, thus hindering the conclusions. However, given the difficulty of detecting disease in patients with early BR of PCa, we believe cfDNA fragmentation could serve as a biomarker to identify oligometastatic patients with low disease burden during follow-up. Further research with larger cohorts is warranted to validate these results and explore the clinical utility of cfDNA fragmentation analysis in this context.

Conclusion

This prospective study demonstrated a statistically significant negative correlation between cfDNA fragmentation patterns and PSMA PET/CT volumetric parameters in localized prostate cancer patients with early biochemical recurrence, consistent with previous research. While this work is limited by its sample size, these findings underscore the potential of liquid biopsy as a biomarker and a complementary tool to PSMA PET/CT to assess disease progression during the follow-up of these patients.

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Author contributions

All authors (GA, MF, JM, LM, MJR, KQ, RP, MIT, FLR, DF, AA, LI, PP) contributed to the study conception and design. PP and LI coordinated and supervised the study. Material preparation, data collection and analysis were performed by GA, MF, PP and LI. The first draft of the manuscript was written by GA and MF and all other authors (JM, LM, MJR, KQ, RP, MIT, FLR, DF, AA, LI, PP) commented on previous versions of the manuscript. All authors (GA, MF, JM, LM, MJR, KQ, RP, MIT, FLR, DF, AA, LI, PP) read and approved the final manuscript.

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Availability of data and materials

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study methodology conformed to the standard set by the Declaration of Helsinki and was approved by the Clinical Research Ethics Committee of the Hospital Clínic Barcelona (HCB/2022/0542). All participants provided written informed consent (HCB/2013/8753) before being included in this study.

Consent for publication

Written informed consent was obtained from the patient for publication of this study and accompanying images.

Competing interests

The authors have no relevant financial or non-financial interests to disclose.

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